R13

Code No: 114CS

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Write the recurrence relation for divide and conquer. [2]
b) List out the drawbacks of merge sort algorithm. [3]
c) Describe and/or graph. [2]
d) Explain briefly game trees. [3]
e) List the applications of Dynamic programming. [2]
f) Define feasible solution, optimal solution and objective function. [3]
g) Differentiate live node and dead node. [2]
h) Define solution state and answer state. [3]
i) What is decision problem? [2]
j) Define class NP. [3]

PART-B

(50 Marks)

2.a) Define algorithm. Write about algorithm design techniques.
b) Describe Strassen's matrix multiplication. Perform the multiplication operation on following two matrices using Strasen's technique. [5+5]

$$\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix} \times \begin{bmatrix} 2 & 7 \\ 8 & 3 \end{bmatrix}$$

OR

3.a) What is amortized analysis? Explain in detail about techniques used in amortized analysis with help of dynamic array example.
b) Show how quick sort sorts the following sequences of keys 65, 70, 75, 80, 85, 60, 55, 50, 45 solve the recurrence relation of quick sort using substitution method. [5+5]

4.a) Write the pseudo codes for various operations used by disjoint set.
b) Define articulation point. Write the algorithm to find articulation point. [5+5]

OR

5.a) Explain the graph traversal method with help of following graph shown in figure 1.

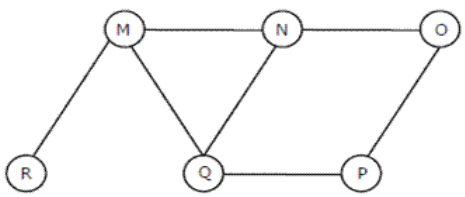


Figure: 1

b) Write and explain the FIND algorithm for collapse rule with example. [5+5]

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6.a) A thief enters a house for robbing thief can carry a maximum weight of 6kg into his bag. There are 3 items in the house with weights (w_1, w_2, w_3) = (2, 3, 4) and profits (p_1, p_2, p_3) = (1, 2, 5) what items should thief can take will get the maximum profit? He either takes or leaves the item.

b) Describe in detail job sequencing with deadlines problem. Let n= 4, (P_1, P_2, P_3) = (100, 10, 15, 27) and (d_1, d_2, d_3, d_4) = (2, 1, 2, 1) find the optimal solution. [5+5]

OR

7. Explain prim's and krushkal's algorithm shown in figure 2 with the following example. [10]

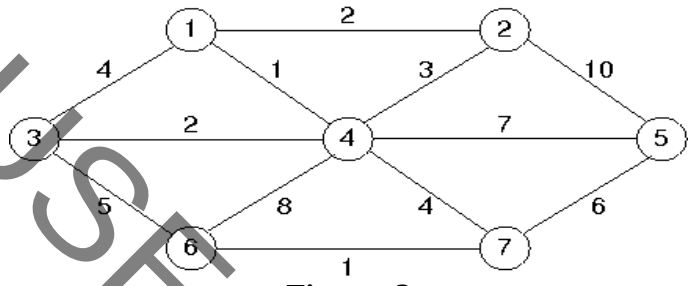


Figure: 2

8. What is n-queen problem? Describe 4-queen problem using back tracking. [10]

OR

9. Find the minimum cost of Tour (TSP) for the following adjacency matrix shown in figure 3 using Branch and Bound. [10]

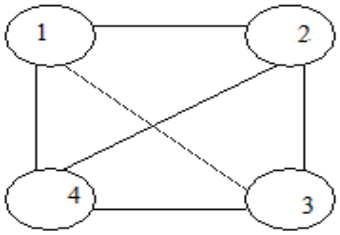


Figure 3

$$\begin{bmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{bmatrix}$$

10.a) Write the Non deterministic algorithm for sorting.

b) Explain CNF-SAT Problem. [5+5]

OR

11.a) Explain about cook's theorem.

b) Explain the P, NP-Hard and NP-complete classes? Give the relation between them. [5+5]

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